

Install Python module

This tutorial is to install the Python driver library of the robotic arm into the virtual machine system.

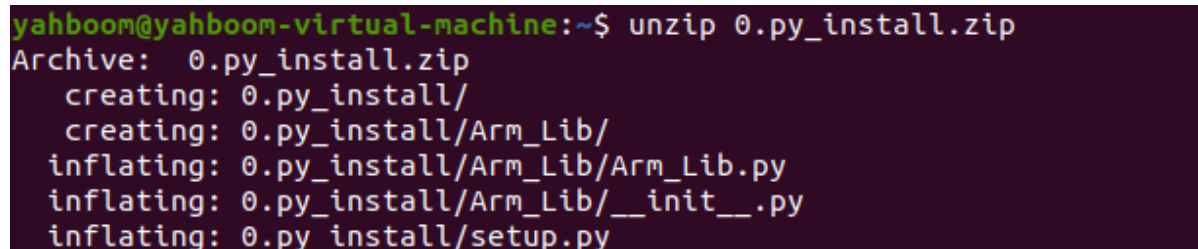
If you use the virtual machine file provided by Yahboom, the Arm_Lib module is already installed in it and there is no need to install it again.

1. Transfer files

Copy the py_install.zip file in the current folder to your virtual machine user directory, and enter the command to decompress it.

```
unzip 0.py_install.tar.gz
```

After successful decompression, the system will display the following interface.



```
yahboom@yahboom-virtual-machine:~$ unzip 0.py_install.zip
Archive:  0.py_install.zip
  creating: 0.py_install/
  creating: 0.py_install/Arm_Lib/
  inflating: 0.py_install/Arm_Lib/Arm_Lib.py
  inflating: 0.py_install/Arm_Lib/__init__.py
  inflating: 0.py_install/setup.py
```

2. Install python library

2.1 Enter the 0.py_install directory.

Note: If the transmission path is different, please use the actual path to execute. There is an Arm_Lib folder and a setup.py file under this path.

```
cd 0.py_install/
```

2.2 Enter the following command to install the python library, enter the password and press **【Enter】** key to confirm.

The system will prompt that the installation is complete and display the version number.

```
sudo python3 setup.py install
```

```
yahboom@yahboom-virtual-machine: ~/Dofbot/0.py_install
copying build/lib/Arm_Lib/__init__.py -> build/bdist.linux-x86_64/egg/Arm_Lib
byte-compiling build/bdist.linux-x86_64/egg/Arm_Lib/Arm_Lib.py to Arm_Lib.cpython-38.pyc
byte-compiling build/bdist.linux-x86_64/egg/Arm_Lib/__init__.py to __init__.cpython-38.pyc
creating build/bdist.linux-x86_64/egg/EGG-INFO
copying Arm_Lib.egg-info/PKG-INFO -> build/bdist.linux-x86_64/egg/EGG-INFO
copying Arm_Lib.egg-info/SOURCES.txt -> build/bdist.linux-x86_64/egg/EGG-INFO
copying Arm_Lib.egg-info/dependency_links.txt -> build/bdist.linux-x86_64/egg/EGG-INFO
copying Arm_Lib.egg-info/top_level.txt -> build/bdist.linux-x86_64/egg/EGG-INFO
zip_safe flag not set; analyzing archive contents...
creating 'dist/Arm_Lib-0.0.5-py3.8.egg' and adding 'build/bdist.linux-x86_64/egg'
to it
removing 'build/bdist.linux-x86_64/egg' (and everything under it)
Processing Arm_Lib-0.0.5-py3.8.egg
Removing /usr/local/lib/python3.8/dist-packages/Arm_Lib-0.0.5-py3.8.egg
Copying Arm_Lib-0.0.5-py3.8.egg to /usr/local/lib/python3.8/dist-packages
Arm-Lib 0.0.5 is already the active version in easy-install.pth

Installed /usr/local/lib/python3.8/dist-packages/Arm_Lib-0.0.5-py3.8.egg
Processing dependencies for Arm-Lib==0.0.5
Finished processing dependencies for Arm-Lib==0.0.5
```

2.3 Input the following command to view the module list of pip3.

```
pip3 list
```

```
-----
absl-py                1.4.0
actionlib              1.14.0
angles                 1.9.13
anyio                  3.7.0
apturl                 0.5.2
argon2-cffi            21.3.0
argon2-cffi-bindings  21.2.0
Arm-Lib                0.0.5
arrow                  1.2.3
astroid                2.15.5
asttokens              2.2.1
async-lru              2.0.2
attrs                  19.3.0
autobahn               17.10.1
Automat                0.8.0
autopep8               2.0.2
Babel                  2.12.1
backcall               0.2.0
baidu-aip              4.16.11
bcrypt                 3.1.7
beautifulsoup4         4.12.2
bleach                 6.0.0
```

3. How to uninstall Arm_Lib library

This step only occurs when Arm_Lib is no longer used.

When prompted whether to delete, enter [Y] and press [Enter] to confirm the uninstall.

```
sudo pip3 uninstall Arm_Lib
```

```

WARNING: pip is being invoked by an old script wrapper. This will fail in a future
version of pip.
Please see https://github.com/pypa/pip/issues/5599 for advice on fixing the underlying
issue.
To avoid this problem you can invoke Python with '-m pip' instead of running pip
directly.
WARNING: The directory '/home/jetson/.cache/pip' or its parent directory is not
owned or is not writable by the current user. The cache has been disabled. Check
the permissions and owner of that directory. If executing pip with sudo, you may
want sudo's -H flag.
Found existing installation: Arm-Lib 0.0.4
Uninstalling Arm-Lib-0.0.4:
  Would remove:
    /usr/local/lib/python3.6/dist-packages/Arm_Lib-0.0.4-py3.6.egg
Proceed (y/n)? y
  Successfully uninstalled Arm-Lib-0.0.4

```

4. Arm_Lib library API

- **Arm_serial_servo_write(self, id, angle, time)**

Control action group operation. 0: Stop 1: Single run 2: Run in loop

- **Arm_Action_Study(self)**

Record the current action once in study mode

- **Arm_Button_Mode(self, mode)**

#Set the mode of K1 button, 0: default mode 1: study mode

- **Arm_Buzzer_Off(self)**

Close buzzer

- **Arm_Buzzer_On(self, delay=255)**

Open the buzzer, delay defaults to 0xff, which control buzzer keeps beeping.

delay=1~50. The buzzer turns off automatically after 100 milliseconds, and the maximum delay time is 5 seconds.

- **Arm_Clear_Action(self)**

#Clear action

- **Arm_PWM_servo_write(self, id, angle)**

Control PWM servo.

id:1-6 (0 means control all servos). angle: 0-180

- **Arm_Product_Select(self, index)**

Set the current product color 1~6, and the RGB light will turn on corresponding color.

- **Arm_RGB_set(self, red, green, blue)**

Control RGB light color

- **Arm_Read_Action_Num(self)**

Read the number of saved action groups 0

- **Arm_get_hardversion(self)**

Read the hardware version number

- **Arm_ping_servo(self, id)**

Read the servo status.

In normally, it will return 0xda; If it didn't read data, it will return 0x00; other values means servo errors.

- **Arm_reset(self)**

#Restart drive board

- **Arm_serial_servo_read(self, id)**

#Read the specified servo angle

id: 1-6 returns 0-180. When read error, it will return None

- **Arm_serial_servo_read_any(self, id)**

Read bus servo angle

id: 1-250

Return value: 0-180

- **Arm_serial_servo_write(self, id, angle, time)**

Control bus servo

id: 1-6 (0 is to send 6 servos)

angle: 0-180°

time: servo rotation time

- **Arm_serial_servo_write6(self, s1, s2, s3, s4, s5, s6, time)**

Control 6 bus servos at the same time

S1~S5 angle:0-180°

S6 angle:0-270°

time: servo rotation time

- **Arm_serial_servo_write6_array(self, joints, time)**

joints: an array, which stores the angle data of 6 servos

time: servo rotation time

- **Arm_serial_servo_write_any(self, id, angle, time)**

Control any bus servo

id: 1-250 (0 is broadcast)

angle: 0-180°

time: servo rotation time

- **Arm_serial_servo_write_offset_state(self)**

Read the status of the position offset of the set bus servo,

0 means no corresponding servo ID, 1 means success, 2 means set failure or it is out of range.

- **Arm_serial_servo_write_offset_switch(self, id)**

Set the bus servo mid-position offset.

id: 1-6 servo

0: Restore the factory default median

- **Arm_serial_set_id(self, id)**
Set the number of the bus servo on expansion board
- **Arm_serial_set_torque(self, onoff)**

Control torque of all servo

1: Turn on the torque

0: Turn off the torque (we can manually change the angle of the servo)